Fused determiner-heads

Fragment 1 — 2026-05-07

This fragment tests the local end of the project: can a CGEL-style category / function distinction be stated as a small constraint on labelled trees? The target is the familiar contrast between an NP with separate daughters in Determiner and Head functions, and an NP in which one determiner-function phrase realizes both functions.

1. Phenomenon

CGEL treats examples such as *three*, *some*, and *many* in independent NP use as cases of fused determiner-heads. The fusion is stated over functions: the same daughter fills Determiner and Head. That daughter is not a bare determiner. In the ordinary simple cases it is a DP; in genitive cases it can be a genitive NP.

(1) *Pat invited two students; I invited three.*

(2) *Some of the reports were late, but many arrived on time.*

(3) *Kim's proposal was stronger than Pat's.*

The descriptive problem is small but useful. In *two students*, the DP *two* fills the determiner function (Det), and the nominal *students* fills Head. In independent *three*, the relevant DP fills both functions. In *Pat's*, a genitive NP does the same work. A fragment should therefore distinguish ordinary determiner-plus-Head NPs from fused determiner-head NPs without recasting the fused function pair as a lexical-category claim.

2. Vocabulary

Use the shared relations $\lhd$ and $\lhd^*$, and the function relations Head and Det. Add one local predicate for the phrase types that can fill Determiner function in this draft:

(4) $\mathrm{DetFiller}(x) \equiv_{\mathrm{def}} \mathrm{DP}(x) \vee \mathrm{GenNP}(x)$
x is a DP or a genitive NP eligible for Determiner function

The predicate GenNP is intentionally provisional: a fuller fragment should say whether it is a feature-bearing subtype of NP, a separate abbreviation, or a definable configuration. For now it keeps the category/function issue visible.

3. Specification

First define the fused function-pair relation:

(5) $\mathrm{FusedDetHead}(d, n) \equiv_{\mathrm{def}} n \lhd d \wedge \mathrm{Det}(n, d) \wedge \mathrm{Head}(n, d).$
d is both Determiner and Head of n

An ordinary NP with separate Determiner and nominal Head daughters can be characterised as follows:

(6) $$\begin{aligned}\mathrm{CanonicalDetHeadNP}(n) \equiv_{\mathrm{def}} \mathrm{NP}(n) \wedge (\exists d)(\exists h)[&n \lhd d \wedge n \lhd h \wedge d \neq h \\ &\wedge \mathrm{Det}(n, d) \wedge \mathrm{DetFiller}(d) \wedge \mathrm{Head}(n, h) \wedge \mathrm{Nom}(h)].\end{aligned}$$
an NP with a determiner-function phrase and a distinct nominal Head

The fused determiner-head configuration is the corresponding case where one determiner-function phrase realizes both functions and no separate nominal Head daughter is present:

(7) $$\begin{aligned}\mathrm{FusedDetHeadNP}(n) \equiv_{\mathrm{def}} \mathrm{NP}(n) \wedge (\exists d)[&\mathrm{DetFiller}(d) \wedge \mathrm{FusedDetHead}(d, n) \\ &\wedge \neg(\exists h)[n \lhd h \wedge h \neq d \wedge \mathrm{Head}(n, h) \wedge \mathrm{Nom}(h)]].\end{aligned}$$
an NP whose DP or genitive NP daughter is both Determiner and Head

For simple cases such as *three*, this is a first-order local definition. Partitive cases add one more daughter, but the fused element is still the DP in the fused function pair:

(8) $\mathrm{PartitiveFusedDetHeadNP}(n) \equiv_{\mathrm{def}} \mathrm{FusedDetHeadNP}(n) \wedge (\exists p)[n \lhd p \wedge \mathrm{Comp}(n, p) \wedge \mathrm{PP}(p)].$
a fused determiner-head NP with an of-PP complement

Nothing in the formula assigns the fused element to a lexical category. It says only that this tree has one daughter bearing two functions. In the ordinary cases that daughter is a DP.

4. Worked instance

Take *I invited three*. Let n be the object NP and d the DP *three*. The intended tree makes d a determiner-function filler and gives d both the Determiner and Head functions in n. There is no distinct nominal Head daughter, so n satisfies FusedDetHeadNP(n).

The contrast with *two students* is immediate. In that NP, the DP *two* fills Det(n, d), while the nominal *students* fills Head(n, h). That NP satisfies CanonicalDetHeadNP(n), not FusedDetHeadNP(n).

The genitive case shows why the analysis cannot be stated as a lexical-category predicate. In *Kim's proposal was stronger than Pat's*, the fused daughter in the second comparison is a genitive NP. It still fills the Determiner and Head functions together.

5. Open questions

(a) The inventory behind DetFiller needs checking. DP and genitive NP are included here because they expose the main category/function contrast, but CGEL's full set of Determiner-function fillers may need a richer definition.

(b) Partitives probably require tighter treatment of the *of*-PP: it is not just any complement PP.

(c) The inventory of non-fused Determiner fillers should be kept separate from the fused-function configuration unless the interaction is explicitly modelled.

(d) CGEL page references and the Payne–Huddleston–Pullum treatment of fused functions need to be checked before this becomes a citable fragment.